
AresSeg: A Controlled Study of CNNs, Transformers, and Foundation Models for Mars Terrain Segmentation

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Abstract

Reliable terrain perception can increase the safety and autonomy of Mars rover traverses, but architecture comparisons are easily confounded by data splits, initialization, and incomplete uncertainty estimates. We present AresSeg, a controlled four-class semantic-segmentation study on AI4Mars Navcam imagery. Thirty learned runs compare U-Net, DeepLabV3+, SegFormer, a small U-Net, and a satellite-pretrained DINOv3 model at three seeds; a parameter-free majority predictor and the Segment Anything Model (SAM) region-oracle serve as references. All systems share a leakage-safe split, ignore-label handling, objective, expert test sets, and a preregistered paired seed/image bootstrap with Holm correction. Conventional systems reach mean Curiosity test mIoU of 0.810–0.824 and all exceed the training-derived majority baseline by at least 0.754 (adjusted $p < 0.001$). We find no corrected evidence that ImageNet initialization improves a tested architecture and no corrected overall or class-specific difference between the configured SegFormer-B2 and CNN systems. The validation-selected U-Net transfers from Curiosity to Opportunity/Spirit with mean mIoU 0.792, a 0.030 drop and 90% interval $[0.725, 0.848]$, thereby satisfying the prespecified robustness rule. DINOv3-SAT obtains the highest mean mIoU (0.836) and significantly exceeds Tiny U-Net by 0.148, whereas SAM’s label-assisted region upper bound does not. These results favor careful representation transfer and compact SegFormer-B0 efficiency over a simple CNN-versus-transformer narrative.

1 Introduction

Mars rovers operate under communication latency, intermittent contact, and strict safety margins. Longer autonomous drives require immediate understanding of the surface ahead: sand may induce slip, embedded bedrock contributes to wheel wear, and large rocks are direct hazards. SPOC linked deep terrain classification to operational rover mobility [14], while AI4Mars made large-scale pixel supervision available by combining crowdsourced labels with expert test annotations [16]. The remaining practical question is not merely whether a network can segment Mars terrain, but which representation transfers reliably under controlled evaluation and rover shift.

This paper asks: *How do convolutional, transformer, and foundation-model systems compare on four-class Mars terrain segmentation; does pretrained initialization help; and does a Curiosity-selected system generalize to Opportunity/Spirit imagery?* We study soil, bedrock, sand, and big rock in grayscale Curiosity Navcam images, with label 255 excluded from both loss and metrics. The study contributes:

1. a camera-scoped, leakage-safe AI4Mars pipeline with reconstructable per-image statistics;

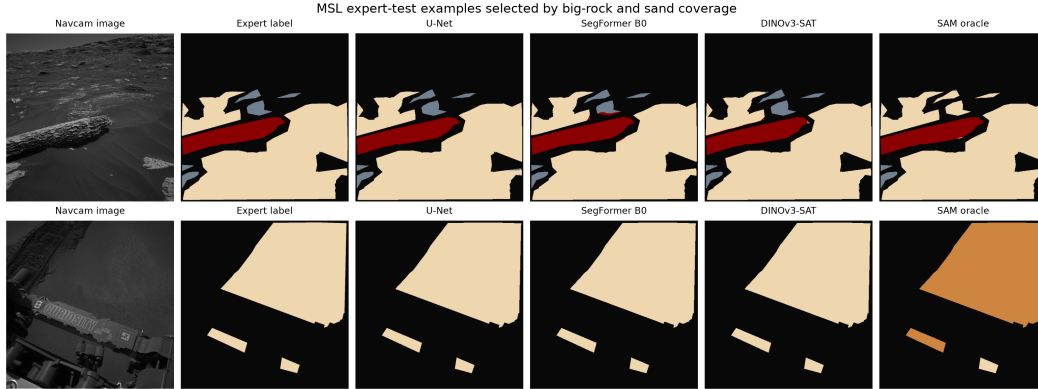


Figure 1: Two Curiosity expert-test scenes selected deterministically for high big-rock (top) and sand (bottom) coverage. SAM is a label-assisted region-oracle upper bound, not a deployable semantic classifier. All colors use one fixed terrain palette; black denotes ignored pixels.

2. a matched three-seed comparison of CNNs, hierarchical transformers, initialization, and two foundation-model references;
3. a held-out Curiosity-to-MER transfer test with preregistered selection and decision rules;
4. a fully sealed analysis trail that records protocol amendments rather than rewriting them.

The visual summary in Figure 1 previews the central empirical pattern: several learned systems produce strong broad-terrain maps, yet rare rocks and class boundaries remain visually and statistically harder than soil or bedrock.

2 Related work

Mars terrain perception. AI4Mars contains rover imagery, crowdsourced training labels, and expert evaluation labels from Curiosity, Spirit, and Opportunity [16]; this work uses its merged-0.6 public release [12]. SPOC previously demonstrated operationally motivated terrain classification for Mars rover missions [14]. MarsSeg explores Mars-specific multi-level feature extraction, but remains a research preprint [8]. Our focus is complementary: controlled comparisons, initialization effects, cross-rover transfer, and paired inference under one protocol.

CNN and transformer segmentation. Fully convolutional prediction established end-to-end dense recognition [9]. U-Net recovers spatial detail through encoder–decoder skip connections [13], whereas DeepLabV3+ combines atrous multi-scale context with a boundary refinement decoder [1]. Residual/ImageNet encoders offer reusable visual priors [2, 5]. Vision Transformers replace local convolution with patch-token attention [3]; SegFormer adapts this idea to efficient hierarchical dense prediction [17]. Because each family changes capacity and optimization as well as inductive bias, our H3 conclusions concern configured systems, not a pure causal effect of attention.

Foundation references. SAM learns class-agnostic object masks at scale [7], but AI4Mars supplies no prompt-to-terrain semantic channel. We therefore use the frozen region-oracle rule only as an upper bound on its proposal partition. DINOv3 is a 2025 technical report, not a peer-reviewed paper [15]. Its SAT-493M representation is nevertheless a useful stress test: it transfers self-supervised features from Earth orbital imagery to ground-level Mars imagery.

3 Problem and hypotheses

For an image $x \in \mathbb{R}^{H \times W}$, replicated to three channels for pretrained encoders, a model f_θ produces four logits per pixel. The prediction is

$$\hat{y}_{ij} = \arg \max_{c \in \{0,1,2,3\}} f_\theta(x)_{ijc}. \quad (1)$$

Pixels satisfying $y = 255$ are invalid and contribute to neither optimization nor sufficient statistics. Class IoU and the primary macro mean are

$$\text{IoU}_c = \frac{\text{TP}_c}{\text{TP}_c + \text{FP}_c + \text{FN}_c}, \quad \text{mIoU} = \frac{1}{|S|} \sum_{c \in S} \text{IoU}_c, \quad (2)$$

where S is the fixed set of classes present in the full evaluation split.

The sealed hypotheses are: H0, no tested deep system beats constant majority; H1, at least one conventional deep system does; H2, pretraining improves at least one backbone-matched system; H3, configured SegFormer-B2 and CNN systems differ overall or by class; H4, the validation-selected system has cross-rover drop < 0.15 and an MER confidence lower bound above MER majority; and H5, DINOv3-SAT or the SAM upper bound exceeds Tiny U-Net. H0 is rejected exactly when H1 is supported. Failure of H2 or H3 is not interpreted as proof of equivalence.

4 Method

4.1 Data and leakage controls

The MSL Navcam index contains 16,064 labeled training images and 322 expert-test images. A seed-1414 by-image split yields 12,851 training and 3,213 validation records; no crop or frame crosses partitions. The independent MER expert set contains 204 Opportunity/Spirit images and is never used for fitting or selection. Preflight verifies counts, image-label pairing, split disjointness, the vocabulary $\{0, 1, 2, 3, 255\}$, source checksum, and a path/size index fingerprint. The dataset is not copied into the repository; the public record and expected MD5 are documented.

Images are resized to 512^2 , replicated to RGB, and ImageNet-normalized. Training uses horizontal flip and mild brightness/contrast perturbation; physically implausible vertical flips are disabled. Masks use nearest-neighbor resizing. Median-frequency class weights are clipped to $[0.5, 10]$ and computed only from the training split.

4.2 Systems and optimization

The 30 learned runs comprise Tiny U-Net; U-Net-ResNet34; DeepLabV3+-ResNet50; SegFormer MiT-B0 and B2, each where applicable with ImageNet and random initialization; and DINOv3-SAT ViT-L/16 with a trained segmentation head. The deterministic training-pixel majority predictor selects bedrock (class 1); its full count evidence is sealed in Protocol V5. SAM ViT-B assigns each generated proposal its majority valid target class and defaults uncovered pixels to soil, making it explicitly label-assisted.

Learned systems minimize weighted cross-entropy plus soft Dice loss [11]:

$$\mathcal{L} = \mathcal{L}_{\text{CE}}^w + \left(1 - \frac{2 \sum p g + \epsilon}{\sum p + \sum g + \epsilon} \right), \quad (3)$$

with the validity mask applied to both terms. AdamW uses learning rate 3×10^{-4} , weight decay 10^{-4} , batch size 8, at most 50 epochs, gradient clipping at 1.0, and early stopping patience 10 [10]. The best validation-mIoU checkpoint is evaluated once.

4.3 Statistical protocol and reproducibility

Learned arms use seeds 1414, 1415, and 1416. For each comparison, 10,000 bootstrap replicates resample training seeds and then paired images within seeds. We report percentile 90% intervals and plus-one p -values [4]. Family A/B tests are one-sided; Family C is two-sided and corrects both overall and four per-class tests across both comparisons. Holm correction controls each preregistered family at $\alpha = 0.10$ [6]; H4 uses its threshold-plus-interval rule without a p -value.

Each run stores its resolved config, Git revision, code/config hashes, environment, parameter counts, best validation score, and per-image intersections/unions. Protocol V5 SHA-256 is 8485810bebed7d1396a0e8c065873ca56c6ea65565f3a919898d46509734ae04. Thirty learned runs used approximately 179.1 V100 device-hours (32 GB V100-SXM2), reconstructed from run start IDs and manifest completion timestamps; the median run was 5.37 hours. Checkpoints and raw data remain untracked.

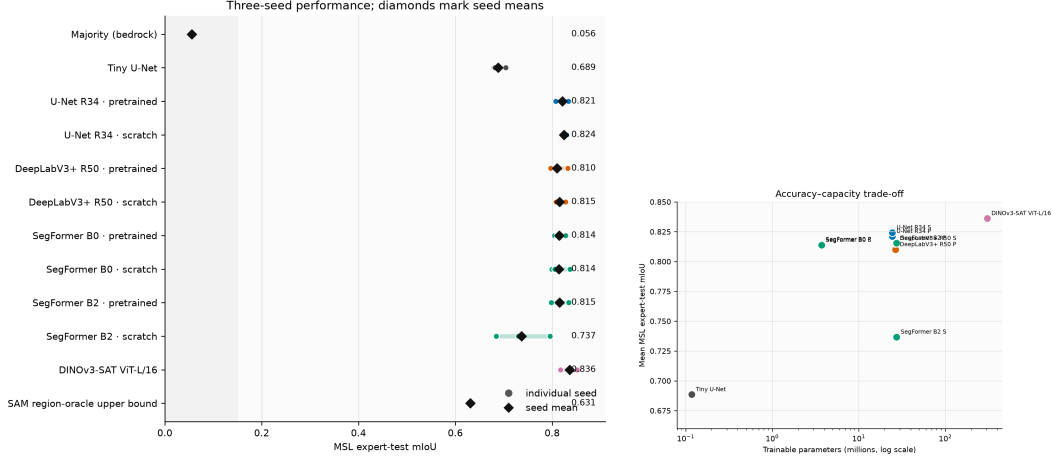


Figure 2: Left: all completed MSL runs; diamonds are seed means and bars span seeds. Right: parameter efficiency for learned systems. SAM and majority are omitted from the capacity panel because their roles are not ordinary trained segmentation models.

Table 1: Expert-test performance. SD is across three training seeds; deterministic references are evaluated once. SAM is a label-assisted region upper bound.

System	Seeds	mIoU (mean $\pm$ SD)	Params
Majority (bedrock)	1	0.056 $\pm$ –	0
Tiny U-Net	3	0.689 $\pm$ 0.014	0.1M
U-Net R34 · pretrained	3	0.821 $\pm$ 0.013	24.4M
U-Net R34 · scratch	3	0.824 $\pm$ 0.005	24.4M
DeepLabV3+ R50 · pretrained	3	0.810 $\pm$ 0.019	26.7M
DeepLabV3+ R50 · scratch	3	0.815 $\pm$ 0.011	26.7M
SegFormer B0 · pretrained	3	0.814 $\pm$ 0.012	3.7M
SegFormer B0 · scratch	3	0.814 $\pm$ 0.020	3.7M
SegFormer B2 · pretrained	3	0.815 $\pm$ 0.018	27.3M
SegFormer B2 · scratch	3	0.737 $\pm$ 0.056	27.3M
DINOv3-SAT ViT-L/16	3	0.836 $\pm$ 0.018	305.5M
SAM region-oracle upper bound	1	0.631 $\pm$ –	93.7M

5 Results

5.1 Overall performance and capacity

Figure 2 and Table 1 summarize the complete inventory. DINOv3-SAT has the highest mean test mIoU (0.836), followed by scratch U-Net (0.824) and pretrained U-Net (0.821). The ranking is descriptive; corrected comparisons below determine the claims. SegFormer B0 is notable for reaching 0.814 with 3.7M parameters, roughly one-seventh of U-Net or DeepLabV3+ and one-eighth of DINOv3-SAT.

5.2 Preregistered hypothesis tests

All conventional systems strongly exceed majority: paired differences are 0.765 for U-Net, 0.754 for DeepLabV3+, and 0.758 for the validation-selected SegFormer-B0, with adjusted $p = 3.0 \times 10^{-4}$ for each. H1 is supported and H0 rejected. In contrast, pretraining effects are -0.003 (U-Net), -0.005

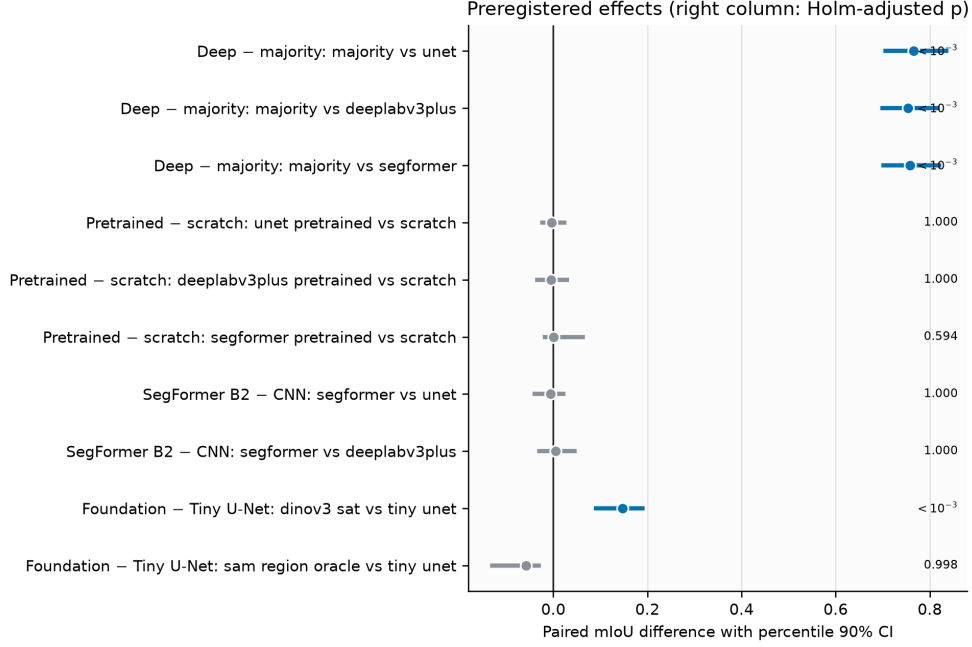


Figure 3: Paired effect estimates. Positive values favor the first named side of each comparison. Blue marks corrected significance; gray marks nonsignificance. Family C per-class tests are reported in the appendix.

Table 2: Final preregistered decisions. “Rejected” means the prespecified support rule was not met, not evidence of model equivalence.

Hypothesis	Decision	Key evidence	Adjusted p
H0	Rejected	H1 supported	–
H1	Supported	$\Delta=0.754\text{--}0.765$	< 0.001
H2	Rejected	all transfer CIs cross 0	≥ 0.594
H3	Rejected	all 10 tests nonsignificant	1.000
H4	Supported	drop=0.030; MER $CI_L=0.725$	rule
H5	Supported	DINOv3 $\Delta=0.148$	< 0.001

(DeepLabV3+), and +0.001 (SegFormer-B0); every 90% interval crosses zero and all adjusted $p \geq 0.594$. H2 is rejected without claiming equivalence.

SegFormer-B2 differs from U-Net by -0.006 mIoU and from DeepLabV3+ by $+0.005$; neither is significant. None of the eight class-specific Family-C tests survives correction (all adjusted $p = 1$), so H3 is rejected. Figure 3 displays all overall effects and asymmetric bootstrap intervals. Table 2 gives the final hypothesis-level decisions.

5.3 Terrain classes, rover shift, and foundation models

The heatmap in Figure 4 shows that strong aggregate scores coexist with weaker and more variable big-rock recognition. The validation-selected pretrained U-Net has mean MSL mIoU 0.821 and mean MER mIoU 0.792. Its drop is only 0.030; the hierarchical MER 90% interval is $[0.725, 0.848]$, whose lower bound is far above MER majority (0.068). H4 is supported, although seed 1416’s MER score (0.714) cautions against a single-run transfer claim.

DINOv3-SAT exceeds Tiny U-Net by 0.148 with interval $[0.091, 0.190]$ and adjusted $p = 2.0 \times 10^{-4}$. SAM’s region-oracle scores only 0.631 and is 0.058 below Tiny U-Net; its one-sided comparison

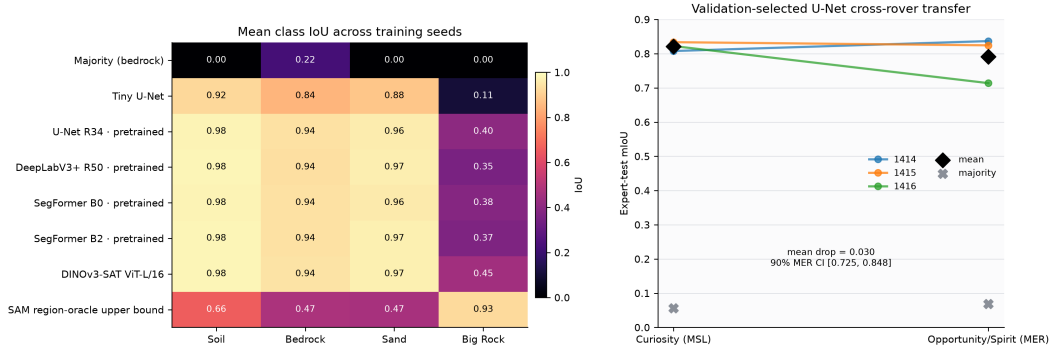


Figure 4: Left: mean terrain-class IoU. Right: paired Curiosity and MER U-Net results, with training-derived majority references. Diamonds denote means.

is not significant. H5 is supported through DINOv3, not through SAM. This distinction matters: strong class-aware features transfer across planets, while generic object partitioning alone does not solve terrain semantics.

6 Discussion

The evidence does not support a simple claim that transformers beat CNNs, or that ImageNet initialization universally helps. U-Net, DeepLabV3+, and both SegFormer sizes occupy a narrow performance band after three seeds; configured-system differences are small relative to image-level and seed uncertainty. Scratch SegFormer-B2 is an exception with high variability, suggesting that the larger transformer is harder to optimize from random initialization, but the pretraining family as formally defined still does not pass correction.

DINOv3-SAT provides the clearest positive transfer result despite the largest viewpoint and planetary domain gap. Its advantage over Tiny U-Net does not establish superiority to all conventional systems, but it does show that large-scale self-supervised satellite features remain useful after adaptation. Conversely, SAM’s target-assisted region upper bound underperforms Tiny U-Net; an autonomous class labeler could only do worse under this proposal partition. Semantic terrain representation, not mask granularity alone, appears to be the decisive ingredient.

Cross-rover transfer is encouraging but uneven. Mean degradation meets a strict bound and MER performance remains far above majority, yet the weakest seed falls more than 0.10 below the other two. Operational use should therefore ensemble or calibrate across conditions rather than deploy a single checkpoint on the basis of mean mIoU. Rare big-rock uncertainty is also more safety relevant than another small improvement on soil.

7 Limitations and broader impacts

The scope is one public dataset release, grayscale Navcam imagery, four coarse classes, and three training seeds. MER shift entangles rover, camera, mission, illumination, and geography, so it does not isolate a causal domain factor. Crowdsourced training labels contain ambiguity even though the test subset requires expert agreement. Macro mIoU weights rare classes equally at aggregation, but rare big-rock pixels still yield wider uncertainty. SAM’s oracle consults targets and is not deployable. DINOv3-SAT is a large preprint model whose compute cost limits onboard use; SegFormer-B0 is the more plausible efficiency candidate. The exploratory MPBA screen was validation-only and not promoted, so it cannot support a gold-test claim.

Safer perception could enable more productive planetary exploration, but segmentation errors may also encourage unsafe autonomous plans. This work is not a flight-qualified navigation system: deployment would require calibrated uncertainty, hardware timing, hazard-aware planning, simulation, Earth-analog field trials, and mission verification. The study contains no personal data and introduces no surveillance application. The merged dataset archive credits NASA/JPL and Zooniverse,

but does not state an explicit license in its bundled metadata; downstream reuse should verify current record terms.

8 Conclusion

Under one leakage-safe, three-seed protocol, conventional CNN and transformer systems all perform strongly and decisively beat majority, but neither ImageNet pretraining nor configured CNN–transformer differences survive multiplicity correction. A validation-selected U-Net meets the pre-registered Curiosity-to-MER robustness rule. DINOv3-SAT yields the highest mean score and supports H5, while SAM proposals alone do not. The most defensible next experiment is therefore a compact DINO-style or SegFormer-B0 representation with explicit uncertainty and rare-rock objectives, evaluated on new rover and lighting strata rather than another unconstrained architecture ranking.

Data and code availability

AI4Mars merged-0.6 is available through Zenodo record 15995036 [12]; the expected archive is approximately 16.2 GB with MD5 daf80a86021253292e6c425f97baa5c6. The public AresSeg repository includes acquisition and integrity instructions, exact configs, protocol seals, compact result stores, manifests, per-image sufficient statistics, analysis code, figures, and a CPU-runnable notebook. Raw data, secrets, checkpoints, and generated prediction rasters are intentionally excluded.

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A Additional qualitative results

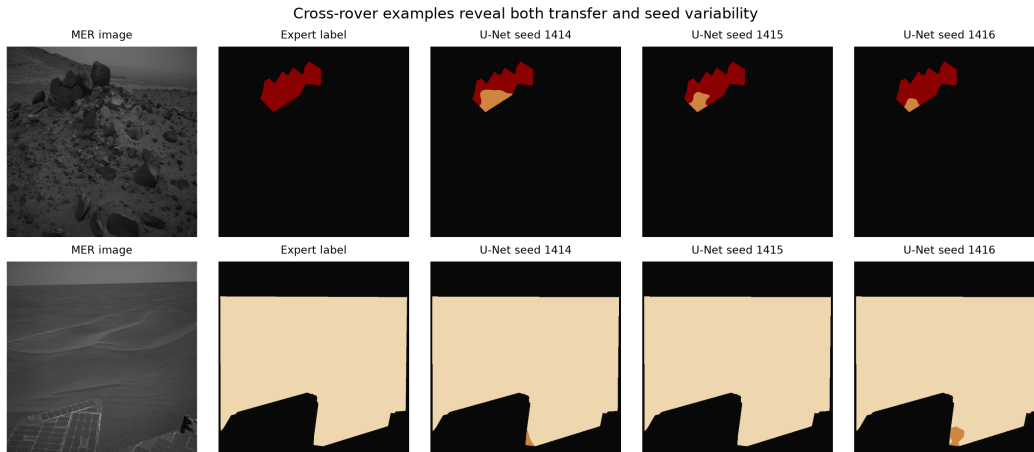


Figure 5: MER scenes selected for high big-rock (top) and sand (bottom) coverage. The three independently trained U-Net checkpoints show where cross-rover predictions are stable and where seed variability enters.

B Exploratory perspective-biased adapter screen

The Mars Perspective-Biased Adapter (MPBA) screen compared native, content-only, static, raw- y , and learned range-cutoff routing for U-Net and SegFormer-B0 on validation seed 1414. Raw- y gained 0.0071 mean mIoU over native across backbones and range-cutoff gained 0.0041, but neither met the joint promotion constraints for big-rock IoU, other-class preservation, latency, and routing use. No variant was evaluated on the gold test, so the experiment remains exploratory.

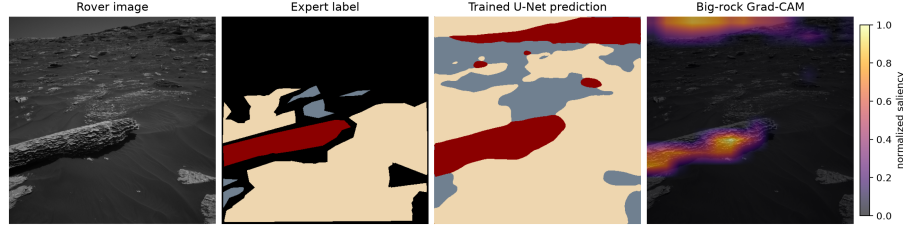
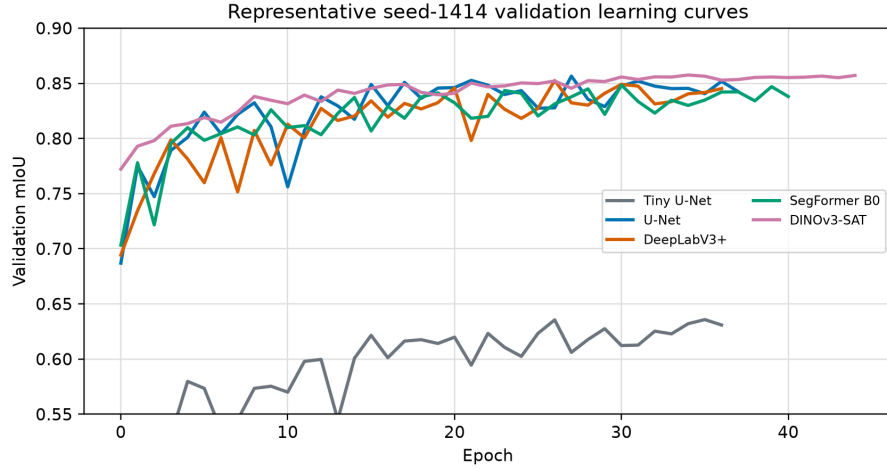


Figure 6: Gradient-based activation visualization for the U-Net big-rock logit. The map is interpretive evidence about spatial sensitivity, not a causal explanation or hypothesis test.



Curves document pipeline convergence; final comparisons use canonical three-seed test artifacts.

Figure 7: Representative seed-1414 validation curves. Final comparisons use the saved best checkpoints and all three seeds.

C Protocol amendments and integrity

The original preregistration and Protocols V2–V4 remain byte-preserved. V4 records the administrative package rename and permits H4 to reuse only checkpoints with one of three exact training commits and the exact shared runtime fingerprint. Before any majority prediction, a fail-closed assertion showed that bedrock, not soil, is the valid training-pixel majority. V5 records the exact counts and changes only the constant predictor from an assumed class to $\arg \max_c n_c$. The failed attempts produced no manifest, prediction, metric, or result row. Canonical CSV and Parquet stores contain 328 rows with no duplicate (`run_id`, `scope`, `stratum`, `metric`) keys.

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Question: Do the main claims made in the abstract and introduction accurately reflect the paper’s contributions and scope?

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Question: Does the paper discuss the limitations of the work performed by the authors?

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Justification: Section 7 discusses dataset, label, seed, domain-shift, compute, oracle-reference, and flight-readiness limitations.

3. Theory assumptions and proofs

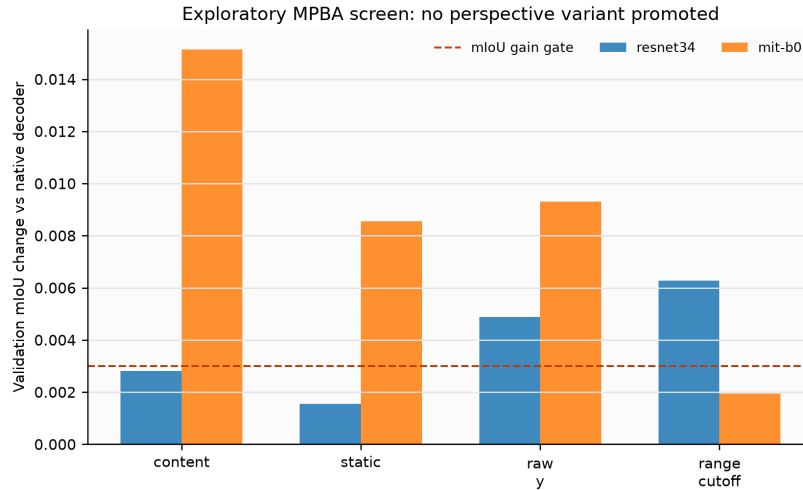


Figure 8: Validation-only MPBA mIoU changes. Crossing the dashed mIoU gate was necessary but not sufficient for promotion; no perspective hypothesis passed the complete rule.

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4. Experimental result reproducibility

Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper?

Answer: [Yes]

Justification: Section 4 gives the data, model, optimization, seed, metric, and inference protocol; sealed configs, manifests, sufficient statistics, and code accompany the paper.

5. Open access to data and code

Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

Answer: [Yes]

Justification: The source repository is public and cites the public Zenodo dataset with acquisition, checksum, layout, and notebook instructions; large raw data and weights are intentionally not duplicated.

6. Experimental setting/details

Question: Does the paper specify all the training and test details necessary to understand the results?

Answer: [Yes]

Justification: Sections 3–4 specify splits, image size, augmentations, objective, optimizer, hyperparameters, checkpoint selection, seeds, and test sets.

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Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

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Justification: Section 4.3 defines the 10,000-replicate seed/image bootstrap, 90% percentile intervals, plus-one p-values, and Holm families; Section 5 reports intervals and adjusted tests.

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Question: For each experiment, does the paper provide sufficient information on the computer resources needed to reproduce the experiments?

Answer: [Yes]

Justification: Section 4.3 reports V100-SXM2 32 GB hardware, 30 learned runs, 179.1 device-hours, and median run time; manifests retain per-run timestamps and environments.

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Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

Answer: [Yes]

Justification: Section 7 discusses scientific productivity, rover safety, the harm of confident perception errors, and required deployment safeguards.

11. **Safeguards**

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Answer: [No]

Justification: All papers, models, NASA/JPL imagery, and the exact Zenodo release are credited, but the bundled merged-0.6 archive metadata does not state an explicit dataset license; Section 7 flags this for downstream verification.

13. **New assets**

Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

Answer: [Yes]

Justification: The repository documents AresSeg code, configs, notebook, compact result stores, protocol seals, generated figures, and exclusions in its README and development plan.

14. **Crowdsourcing and research with human subjects**

Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation?

Answer: [N/A]

Justification: This study recruited no participants. AI4Mars’s prior crowdsourcing is an existing cited public dataset and is not a human-subject experiment conducted here.

15. **Institutional review board approvals or equivalent**

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